Bayesian Data Analysis for Agricultural Science: What Why and How

CANVAS 2026

Day 1: October 31st, 3:00 PM – 5:00 PM

Day 2: November 1st, 9:00 AM – 4:00 PM

Workshop Learning Objectives

Day 1: What and Why

- Understand what we mean by Bayesian data analysis.
- Appreciate why Bayesian data analysis is useful for agronomic research.

Day 2: How

- Learn a process for building statistical models from conceptual models.
- Gain familiarity with computational tools in R and resources for continued learning.
- Complete a mini project on a sample dataset
- Plan/start a Bayesian data analysis project for your own research/dataset.

What do we mean by Bayesian data analysis?

A theoretical framework/way of thinking

A workflow for scientific inquiry

A set of tools (mathematical theory and computational methods)

Why is Bayesian data analysis useful for agronomic research?

Explicitly quantifies uncertainty about inferences based on data

Does not depend on the law of large numbers (good for small/uneven sample sizes)

Open-source tools make Bayesian methods increasingly accessible.

Case Studies

1) DRIVES Network research

2) Soil Quality Index for Northeast Nigeria

3) Corn yield and management in Malawi

Theory basics:

I. Essence of Bayesian inference
II. Illustrate the theoretical framework for Bayesian inference with a simple example.
III. Compare frequentist and Bayesian inference.
IV. Fundamentals of Bayesian computation (MCMC)

I. The essence of Bayesian inference:

1) Make inferences about unknown/unknowable parameters based on data.*

** Not uniquely Bayesian*

unknown/unobservable parameters		**known/observable** data
Disease resistance		Occurrence or severity of disease
Drought resilience		Yields in dry and wet years
Effect of management on soil health		Management history and soil health indicators

The essence of Bayesian inference:

1) Make inferences about unknown/unknowable parameters based on data.*

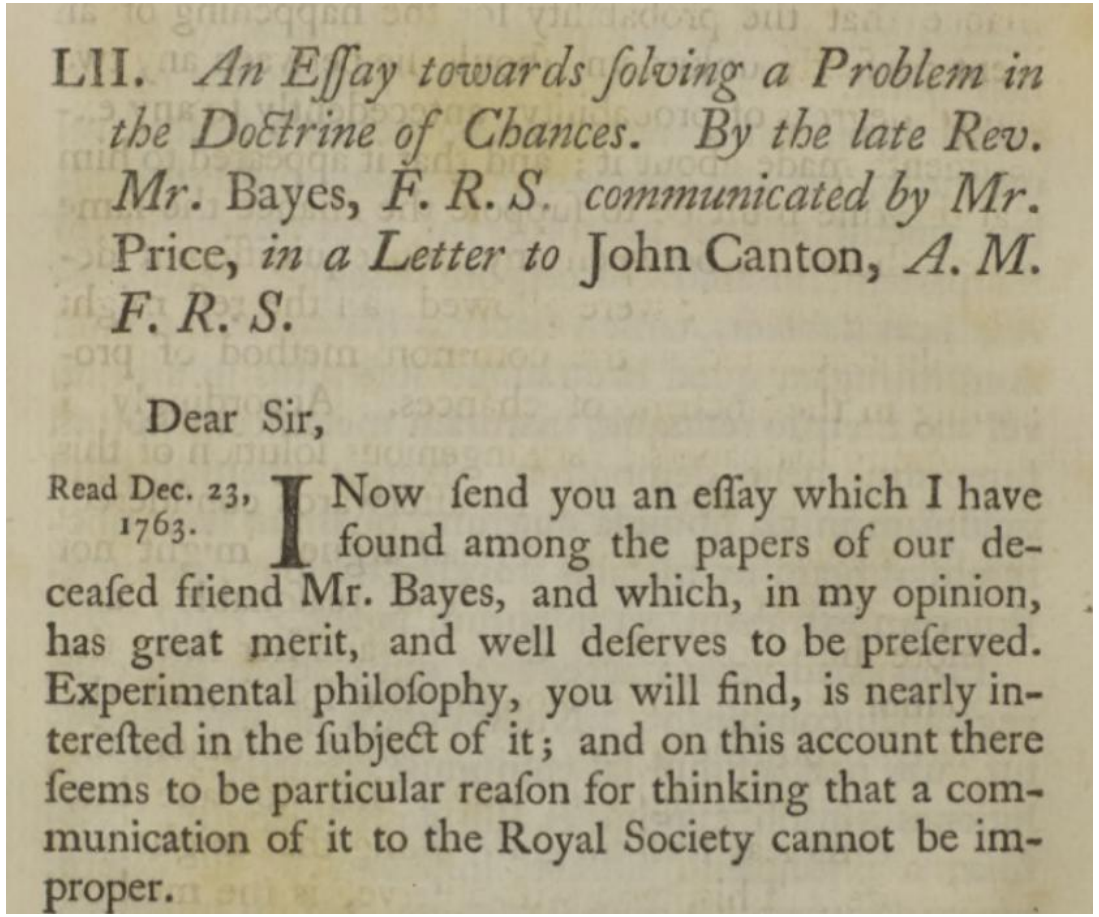
** Not uniquely Bayesian*

2) Use **data** to update our prior understanding about **unknown/unknowable parameters**.

Bayes Rule:

$$P(\boldsymbol{\theta}|\boldsymbol{D}) = \frac{P(\boldsymbol{D}|\boldsymbol{\theta})P(\boldsymbol{\theta})}{P(\boldsymbol{D})}$$

Bayes' Theorem was first published in 1763



LII. *An Eſſay towards ſolving a Problem in the Doctrine of Chances. By the late Rev. Mr.* Bayes, *F. R. S. communicated by Mr.* Price, *in a Letter to* John Canton, *A. M. F. R. S.*

Dear Sir,

Read Dec. 23, 1763. I Now ſend you an eſſay which I have found among the papers of our deceaſed friend Mr. Bayes, and which, in my opinion, has great merit, and well deſerves to be preſerved. Experimental philoſophy, you will find, is nearly intereſted in the ſubject of it; and on this account there ſeems to be particular reaſon for thinking that a communication of it to the Royal Society cannot be improper.

"Bayesian inference" came later

Bayesian Analysis (2006) 1, Number 1, pp. 1–40

When Did Bayesian Inference Become "Bayesian"?

Stephen E. Fienberg*

Abstract. While Bayes' theorem has a 250-year history, and the method of inverse probability that flowed from it dominated statistical thinking into the twentieth century, the adjective "Bayesian" was not part of the statistical lexicon until relatively recently. This paper provides an overview of key Bayesian developments, beginning with Bayes' posthumously published 1763 paper and continuing up through approximately 1970, including the period of time when "Bayesian" emerged as the label of choice for those who advocated Bayesian methods.

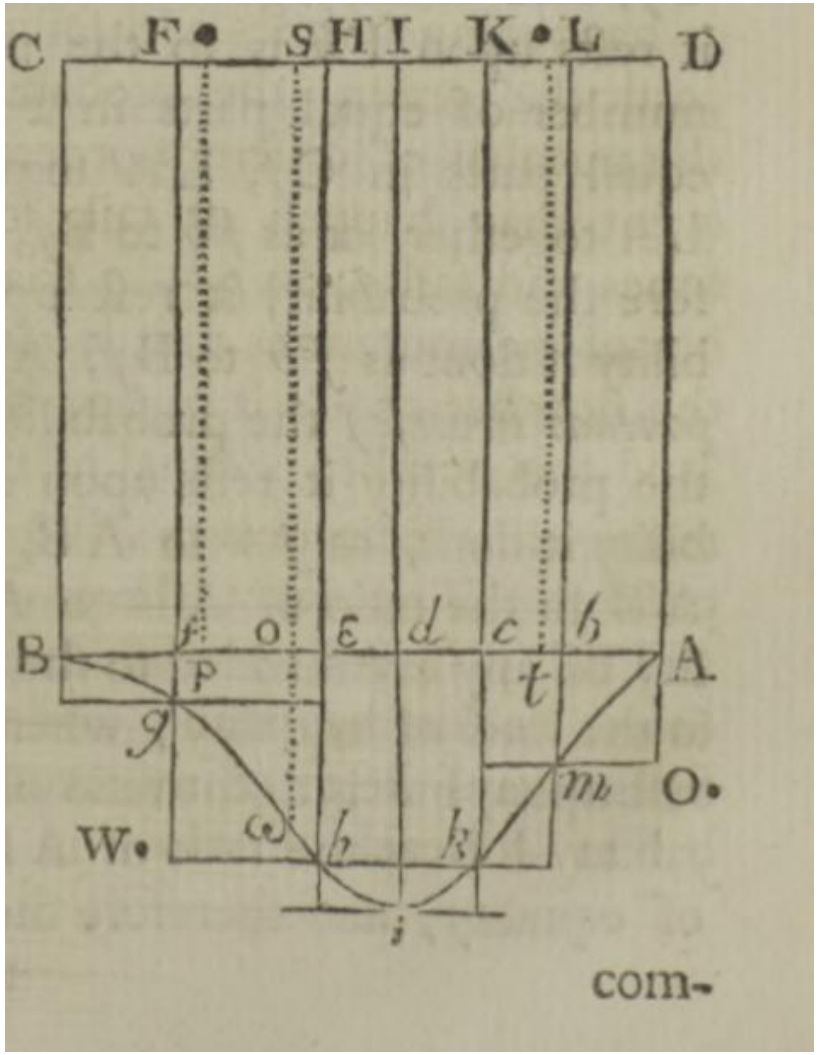
Keywords: Bayes' Theorem; Classical statistical methods; Frequentist methods; Inverse probability; Neo-Bayesian revival; Stigler's Law of Eponymy; Subjective probability.

https://projecteuclid.org/journals/bayesian-analysis/volume-1/issue-1/When-did-Bayesian-inference-become-Bayesian/10.1214/06-BA101.pdf

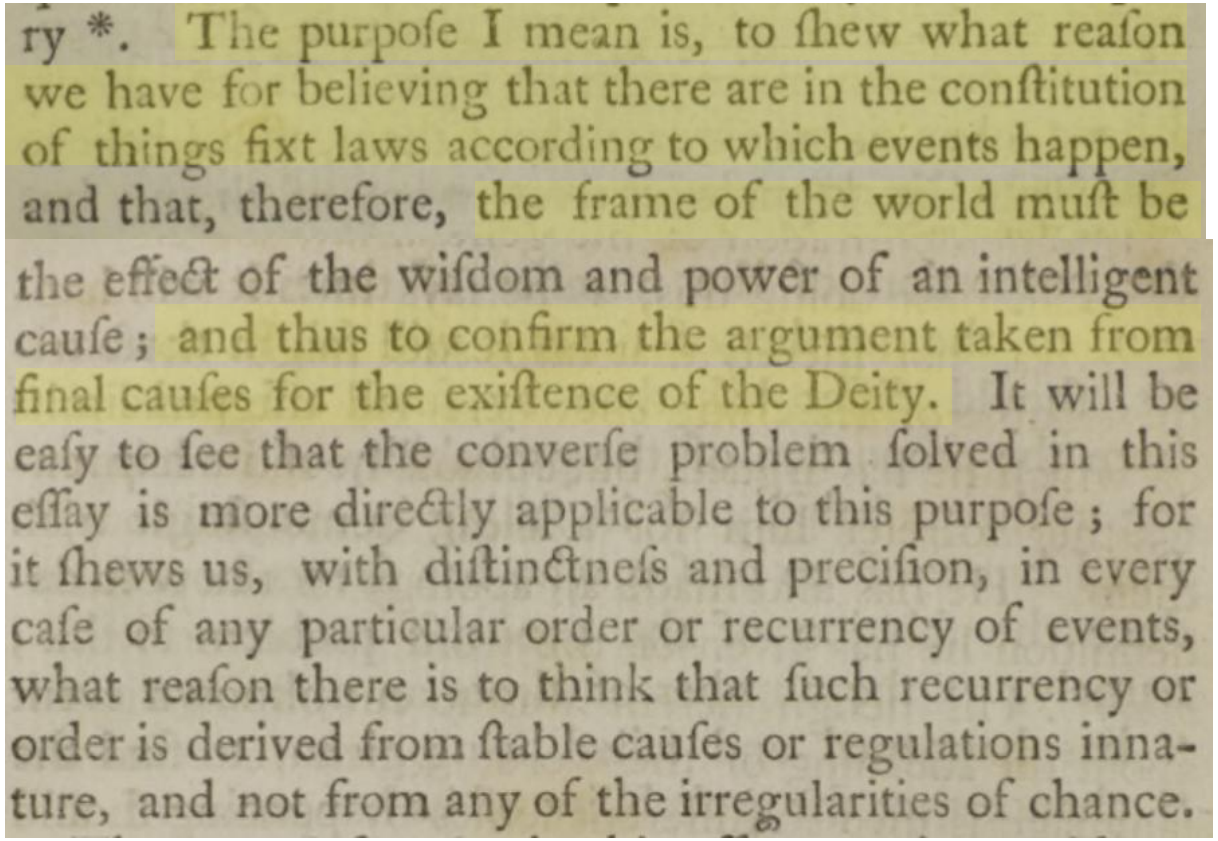
Bayes' main innovation was modeling binary data driven by an underlying continuous parameter → balls rolling on a table

SECTION II.

Poſtulate. 1. I Suppoſe the ſquare table or plane ABCD to be ſo made and levelled, that if either of the balls *o* or W be thrown upon it, there ſhall be the ſame probability that it reſts upon any one equal part of the plane as another, and that it muſt neceſſarily reſt ſomewhere upon it.



Fun fact: Bayes' friend Richard Price thought Bayes' theories confirmed the existence of God.



ry *. The purpoſe I mean is, to ſhew what reaſon we have for believing that there are in the conſtitution of things fixt laws according to which events happen, and that, therefore, the frame of the world muſt be the effect of the wiſdom and power of an intelligent cauſe; and thus to confirm the argument taken from final cauſes for the exiſtence of the Deity. It will be eaſy to ſee that the converſe problem ſolved in this eſſay is more directly applicable to this purpoſe; for it ſhews us, with diſtinctneſs and preciſion, in every caſe of any particular order or recurrency of events, what reaſon there is to think that ſuch recurrency or order is derived from ſtable cauſes or regulations inna-
ture, and not from any of the irregularities of chance.

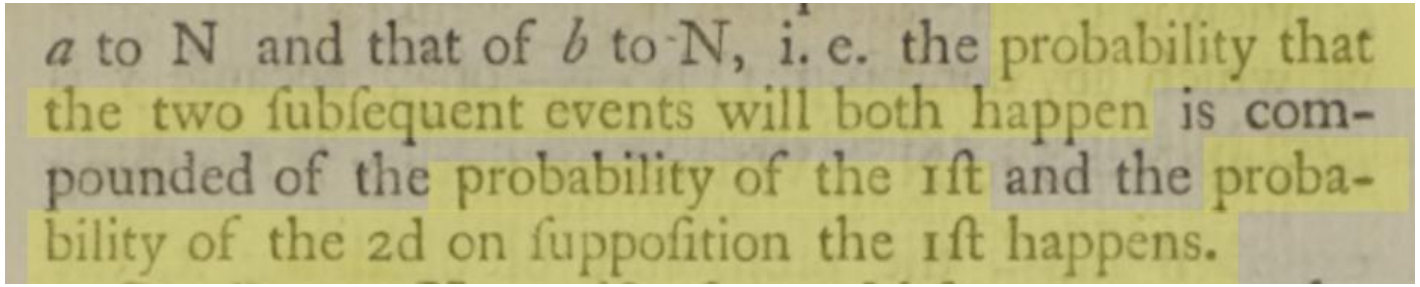
your very humble ſervant,
Newington-Green, Nov. 10, 1763.
Richard Price.

Ccc 3 SEC-

$$P(\boldsymbol{\theta}|\boldsymbol{D}) = \frac{P(\boldsymbol{D}|\boldsymbol{\theta})P(\boldsymbol{\theta})}{P(\boldsymbol{D})}$$

Predict **unknown/unobservable** parameters based on **known/observable** data and prior understanding about parameters.

What we call Bayes' Rule is elementary probability



a to N and that of *b* to N, i. e. the probability that the two ſubſequent events will both happen is compounded of the probability of the 1ſt and the probability of the 2d on ſuppoſition the 1ſt happens.

$$P(A,B) = P(A)\,P(B|A)$$

By the same logic:

$$P(A,B) = P(B)\,P(A|B)$$

Therefore:

Rearrange:

$$P(B)\,P(A|B) = P(A)\,P(B|A)$$

Bayes' Rule

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

$$P(\boldsymbol{\theta}|\boldsymbol{D}) = \frac{P(\boldsymbol{D}|\boldsymbol{\theta})P(\boldsymbol{\theta})}{P(\boldsymbol{D})}$$

Predict **unknown/unobservable** parameters based on **known/observable** data and prior understanding about parameters.

II. Simple linear regression example



Yield = Function(Rainfall, Random)

Data story

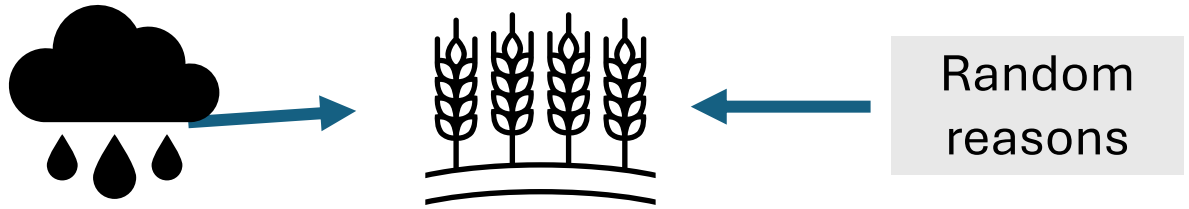
- Rainfed rice is grown **on one farm** under optimal temperature, soil and management.
- Rice is planted and harvested **once per year every year** (no carryover).
- Rice yield is limited by **water availability through rainfall**.
- Rice yield varies randomly from year to year for reasons unrelated to rainfall

Simple linear regression

- Two continuous random variables, **X**, and **Y**
- A change in **X** is associated with a change in **Y**.

$$y = \beta_0 + \beta_1 x + \varepsilon$$

intercept slope error



Rainfall
mm/year

Rainfed rice yield
kg/ha

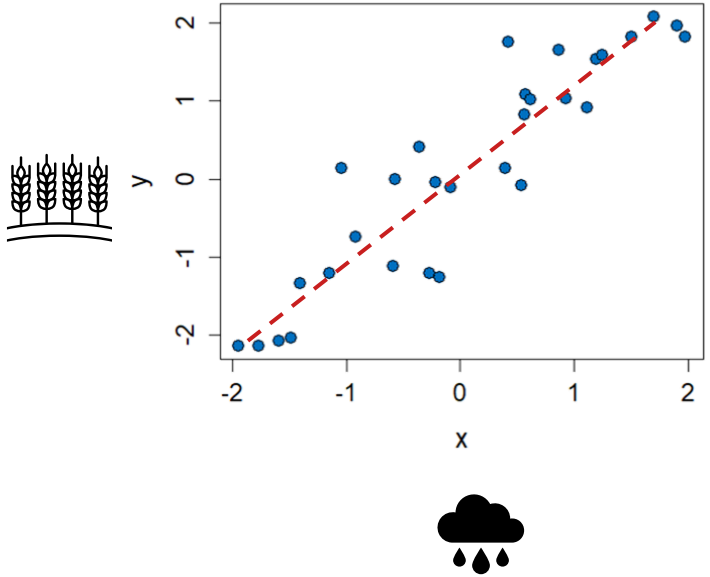
Error has Gaussian noise $\varepsilon \sim \mathcal{N}(0, 1/\tau)$

Yield = Function(Rainfall, Random)

Data processing: *standardize variables*

$$y = \frac{yield - \text{mean}(yield)}{\text{sd}\,(yield)}$$

$$x = \frac{rainfall - \text{mean}(rainfall)}{\text{sd}\,(rainfall)}$$



Precision (τ) is the reciprocal of variance

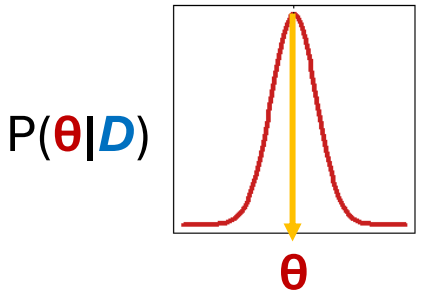
Data: *x* and *y*

Parameters (θ): β_0, β_1, τ

Overview

Frequentist inference

- Focus is on parameter **values**



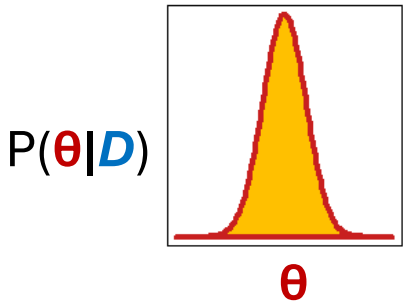
- Assumes **no prior understanding** about parameters

- **Uncertainty** about parameters must be estimated indirectly, using resampling or asymptotic theory.

With weak priors, Bayesian and frequentist methods give the same parameter estimates.

Bayesian inference

- Focus is on the parameter **distributions**



- **Incorporates prior understanding** (or lack thereof).

- **Uncertainty** about parameters is baked into the model

Frequentist Inference

Maximum Likelihood Estimation

Find **values** for each **parameter** that maximize the likelihood of the **data**

Likelihood function: $P(\boldsymbol{D} \mid \boldsymbol{\theta})$

Probability of the **data** given **parameters**

Easily solvable with math! (for simple models)

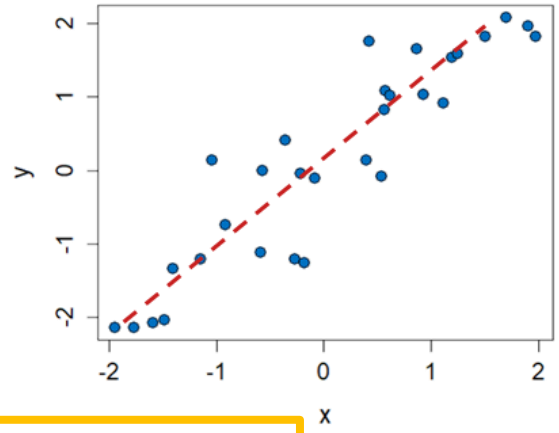
Simple linear regression:

$$\boldsymbol{y} = \beta_0 + \beta_1 \boldsymbol{x} + \boldsymbol{\varepsilon} \qquad \boldsymbol{\varepsilon} \sim \mathcal{N}(0, 1/\tau)$$

$$p(\boldsymbol{y}, \boldsymbol{x} \mid \beta_0, \beta_1, \tau) = \prod_{i=1}^{n} \mathcal{N}(\underbrace{\beta_0 + \beta_1 x_i}_{\text{line}}, \underbrace{1/\tau}_{\text{scatter}})$$

Find the values of β_0, β_1, τ that maximize this function with the observed **x** and **y**.

Equivalent for β_0 and β_1



Ordinary Least Squares:
Find the values of β_0, β_1 that draw the best fit line, minimizing the distance between line and points—i.e., sum of squared residuals

Bayesian Inference

Update our prior understanding about the **distribution** of **parameters** based on the **data.**

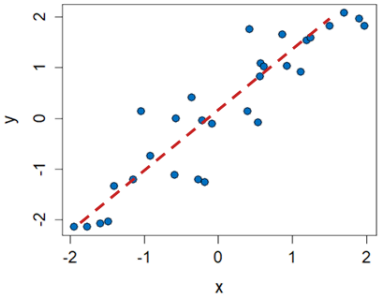
Likelihood of the data given parameters

Prior understanding about parameters

$$\underbrace{P(\boldsymbol{\theta}|\boldsymbol{D})}_{\text{Updated understanding about parameters (posterior)}} = \frac{P(\boldsymbol{D}|\boldsymbol{\theta})P(\boldsymbol{\theta})}{\underbrace{P(\boldsymbol{D})}_{\text{Support over plausible values of the data: Normalizing constant so probability sums to 1}}}$$

Simple linear regression:

$$\boldsymbol{y} = \beta_0 + \beta_1 \boldsymbol{x} + \boldsymbol{\varepsilon} \qquad \boldsymbol{\varepsilon} \sim \mathcal{N}(0, 1/\tau)$$



$$p(\beta_0, \beta_1, \tau \mid \boldsymbol{x}, \boldsymbol{y}) = \frac{p(\boldsymbol{x}, \boldsymbol{y}|\beta_0, \beta_1, \tau)p(\beta_0, \beta_1, \tau)}{p(\boldsymbol{x}, \boldsymbol{y})}$$

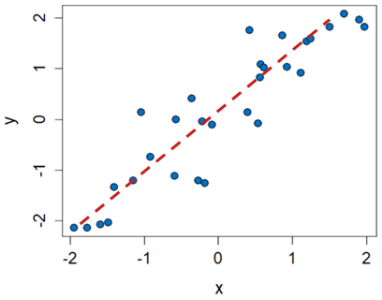
Find **most likely distributions** of β_0, β_1, τ based on observed **x** and **y**.

Bayesian Inference

Update our prior understanding about the **distribution** of **parameters** based on the **data.**

Simple linear regression:

$$\boldsymbol{y} = \beta_0 + \beta_1 \boldsymbol{x} + \boldsymbol{\varepsilon} \qquad \boldsymbol{\varepsilon} \sim \mathcal{N}(0, 1/\tau)$$



Specify prior distributions for parameters

$$P(\boldsymbol{\theta}) \quad \begin{matrix} \beta_0 \sim \\ \beta_1 \sim \\ \tau \sim \end{matrix} \quad \boxed{f(\boldsymbol{\theta}^*)} \quad \begin{matrix} \text{intercept} \\ \text{slope} \\ \text{precision} \end{matrix}$$

hyperparameters

$$p(\beta_0, \beta_1, \tau \mid \boldsymbol{x}, \boldsymbol{y}) = \frac{p(\boldsymbol{x}, \boldsymbol{y} \mid \beta_0, \beta_1, \tau) p(\beta_0, \beta_1, \tau)}{p(\boldsymbol{x}, \boldsymbol{y})}$$

Find **most likely distributions** of β_0, β_1, τ based on observed **x** and **y**.

Choose priors based on:

- Analytical convenience
- How they represent your knowledge (or lack of knowledge) about your system

Bayesian Inference

Update our prior understanding about the **distribution** of **parameters** based on the **data.**

Simple linear regression:

$$y = \beta_0 + \beta_1 x + \varepsilon \qquad \varepsilon \sim \mathcal{N}(0, 1/\tau)$$

Specify prior distributions for parameters

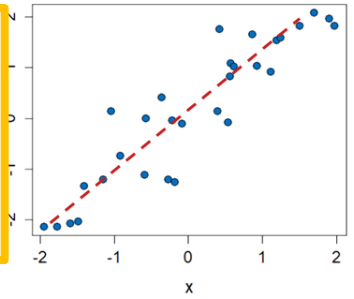
$$P(\boldsymbol{\theta})$$

$$\beta_0 \sim \mathcal{N}(\mu_0, (v_0\tau)^{-1}) \text{ intercept}$$

$$\beta_1 \sim \mathcal{N}(\mu_1, (v_1\tau)^{-1}) \text{ slope}$$

$$\tau \sim \text{Gamma}(a, b) \text{ precision}$$

Prior uncertainty about the regression coefficients must be proportional to prediction uncertainty for the data



$$p(\beta_0, \beta_1, \tau \mid x, y) = \frac{p(x, y|\beta_0, \beta_1, \tau)p(\beta_0, \beta_1, \tau)}{p(x, y)}$$

Find **most likely distributions** of β_0, β_1, τ based on observed **x** and **y**.

Choose priors based on:

- Analytical convenience

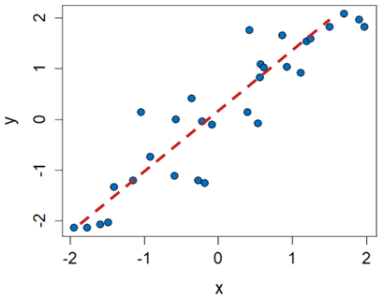
With very specific priors, the posterior can be solved analytically (without computer sampling)

Bayesian Inference

Update our prior understanding about the **distribution** of **parameters** based on the **data.**

Simple linear regression:

$$\boldsymbol{y} = \beta_0 + \beta_1 \boldsymbol{x} + \boldsymbol{\varepsilon} \qquad \boldsymbol{\varepsilon} \sim \mathcal{N}(0, 1/\tau)$$



Specify prior distributions for parameters

$$P(\boldsymbol{\theta}) \quad \begin{aligned} \beta_0 &\sim \mathcal{N}(\mu_0, {\tau_0}^{-1}) \\ \beta_1 &\sim \mathcal{N}(\mu_1, {\tau_1}^{-1}) \\ \tau &\sim \text{Gamma}(a, b) \end{aligned}$$

intercept

slope

precision

Independent hyperparameters

$$p(\beta_0, \beta_1, \tau \mid \boldsymbol{x}, \boldsymbol{y}) = \frac{p(\boldsymbol{x}, \boldsymbol{y} \mid \beta_0, \beta_1, \tau) p(\beta_0, \beta_1, \tau)}{p(\boldsymbol{x}, \boldsymbol{y})}$$

Find **most likely distributions** of β_0, β_1, τ based on observed **x** and **y**.

The posterior can no longer be solved without computer sampling

Recap

- Statistical inference (Bayesian and otherwise) involves estimating unknown or unobservable parameters from observable data (e.g., disease resistance from disease occurrence).
- Frequentist inference differs from Bayesian in that it:
  1. Focuses on parameter values instead of distributions
  2. Estimates parameters solely from the data (no priors)
  3. Is less well-suited to dealing with uncertainty about parameters
  4. Is typically easier and faster to compute.
- Frequentist and Bayesian methods produce similar estimates for parameter values when prior expectations are weak.
- Without computer sampling, Bayesian inference can only be performed on simple models with very restricted priors.

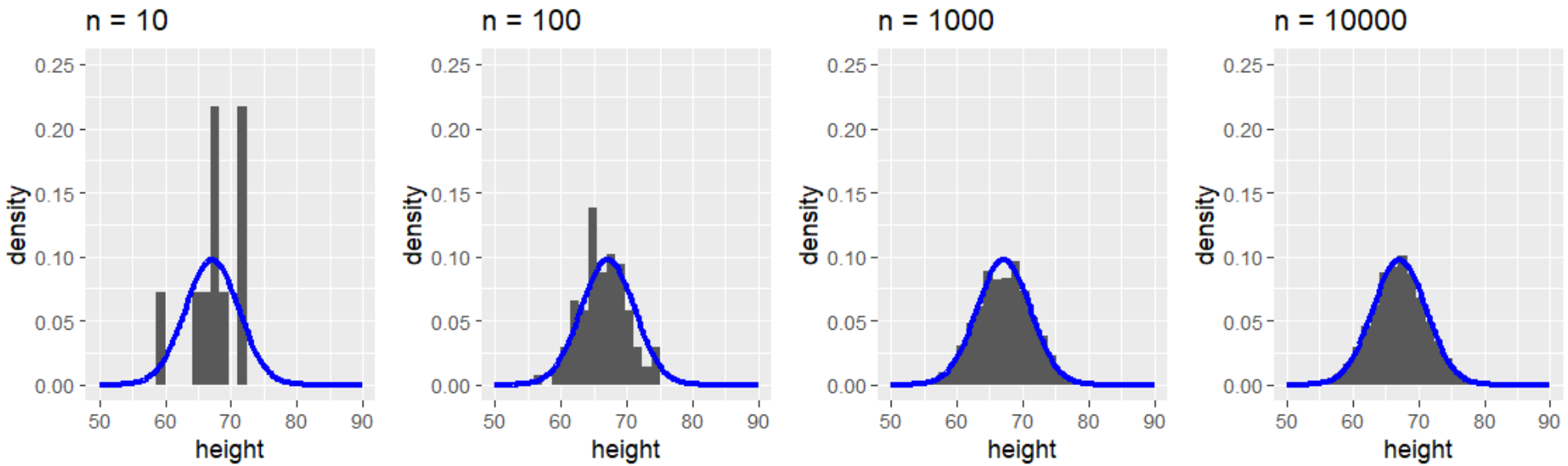
Computers turn an analytical task into a sampling task

$$\text{height} \sim \mathcal{N}(\mu, \sigma)$$

a.k.a. Law of Large Numbers

`rnorm(n, mean, sd)`

More samples → closer to theoretical distribution



Markov chain Monte Carlo (MCMC)

- Family of computer algorithms used for Bayesian data analysis
- Approximate theoretical distributions through iterative sampling
- Gibbs sampling is one type of MCMC algorithm

MCMC algorithms



https://www.geeksforgeeks.org/artificial-intelligence/markov-chain-monte-carlo-mcmc/

Building our model



1. Specify the likelihood

2. Specify priors

3. Sample posterior

$$P(\boldsymbol{\theta}|\boldsymbol{D}) = \frac{P(\boldsymbol{D}|\boldsymbol{\theta})P(\boldsymbol{\theta})}{P(\boldsymbol{D})}$$

Building our model



1. Specify the likelihood

2. Specify priors

3. Sample posterior

$$P(\theta|D) = \frac{P(D|\theta)P(\theta)}{P(D)}$$

Choose a **likelihood function** that **represents our data-generating process**

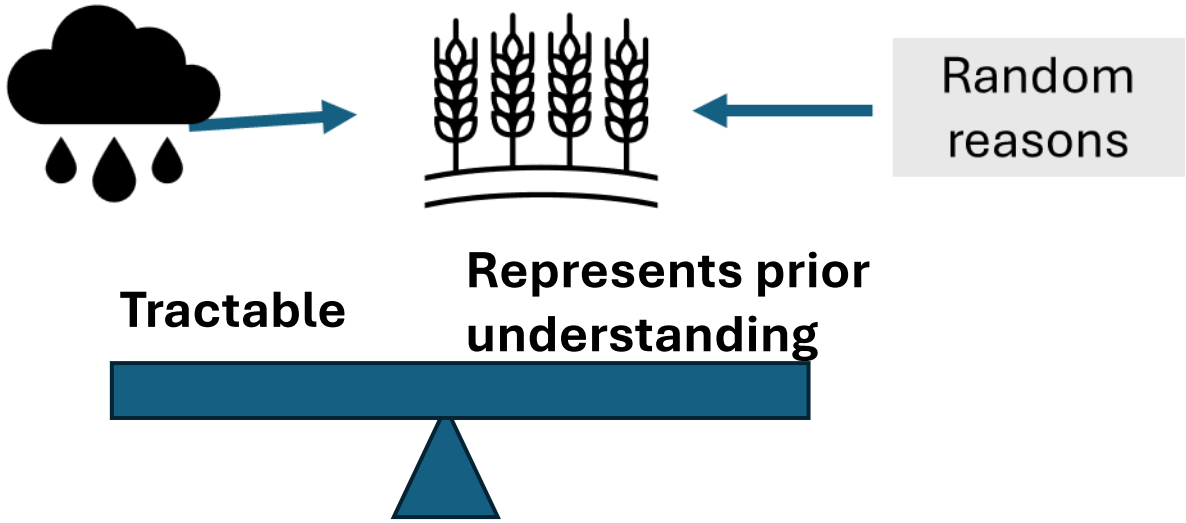
$$y_i \mid x_i, \beta_0, \beta_1, \tau \sim \mathcal{N}(\beta_0 + \beta_1 x_i, 1/\tau)$$

*Yield data (**y**) are generated on a normal distribution with the mean influenced by rainfall (**x**).*

*We can simulate **y** by sampling a normal distribution .*

```
b0  <- 0; b1  <- 1.2; tau <- 3 #"true" parameters
N <- 30 # number of observations of x and y
x <- runif(N,-1,-1) ## random number between -1 and 1
y <- rnorm(N, mean = b0 + b1*x, sd = 1/sqrt(tau))
```

Building our model



1. Specify the likelihood

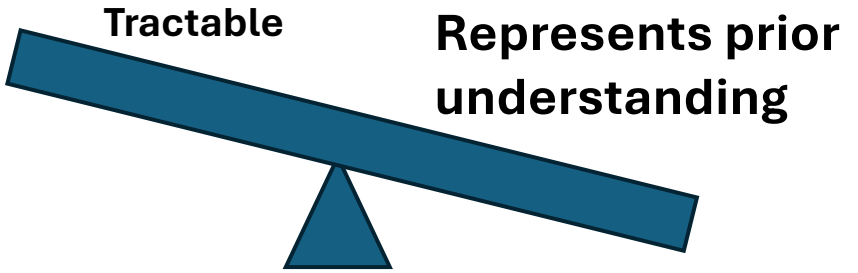
2. Specify priors

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$$P(\boldsymbol{\theta}|\boldsymbol{D}) = \frac{P(\boldsymbol{D}|\boldsymbol{\theta})P(\boldsymbol{\theta})}{P(\boldsymbol{D})}$$

Building our model





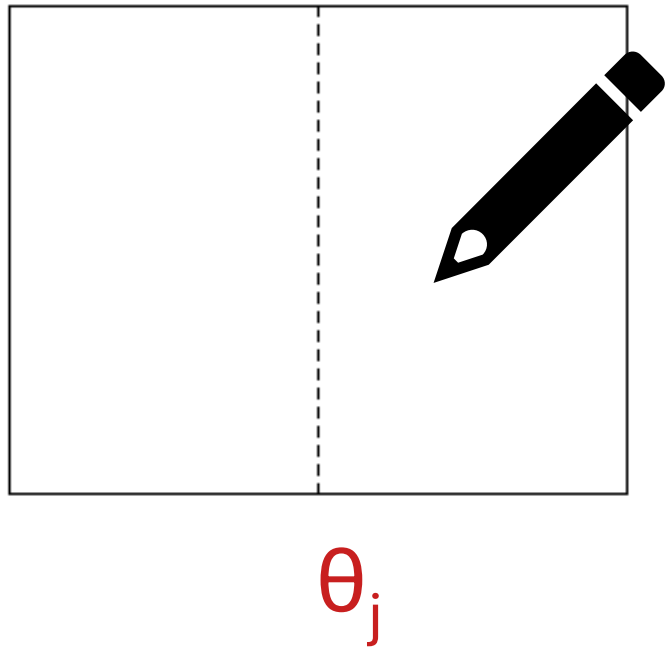
1. Specify the likelihood

2. Specify priors

3. Sample posterior

$$P(\boldsymbol{\theta}|\boldsymbol{D}) = \frac{P(\boldsymbol{D}|\boldsymbol{\theta})P(\boldsymbol{\theta})}{P(\boldsymbol{D})}$$

Plausibility of parameter value $P(\theta_j)$



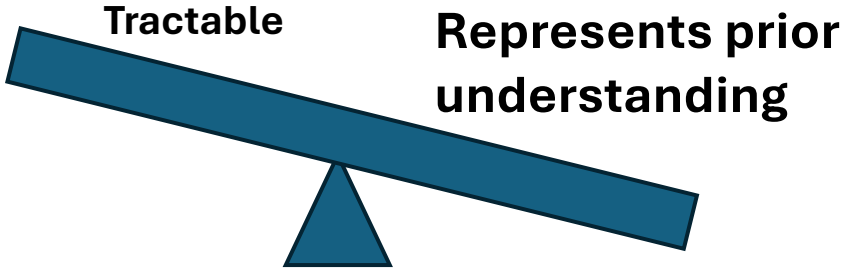
Building our model



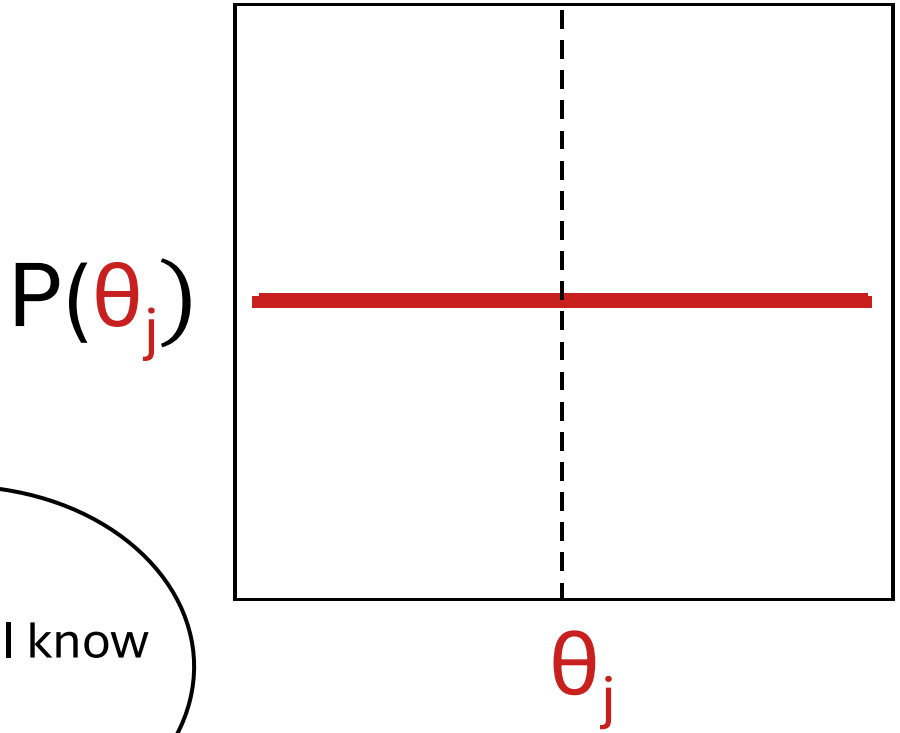
1. Specify the likelihood

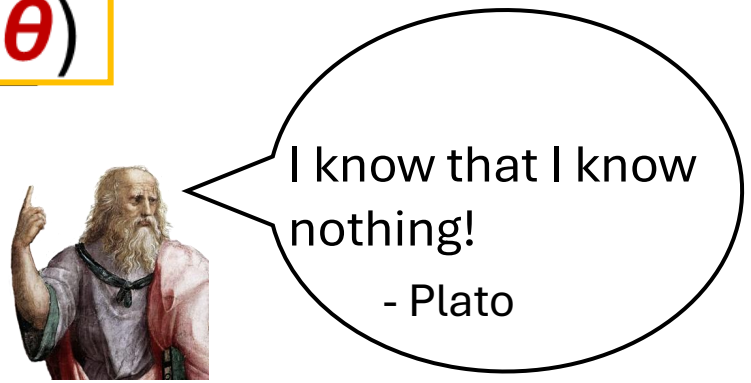
2. Specify priors

3. Sample posterior



$$P(\boldsymbol{\theta}|\boldsymbol{D}) = \frac{P(\boldsymbol{D}|\boldsymbol{\theta})P(\boldsymbol{\theta})}{P(\boldsymbol{D})}$$





Building our model



1. Specify the likelihood

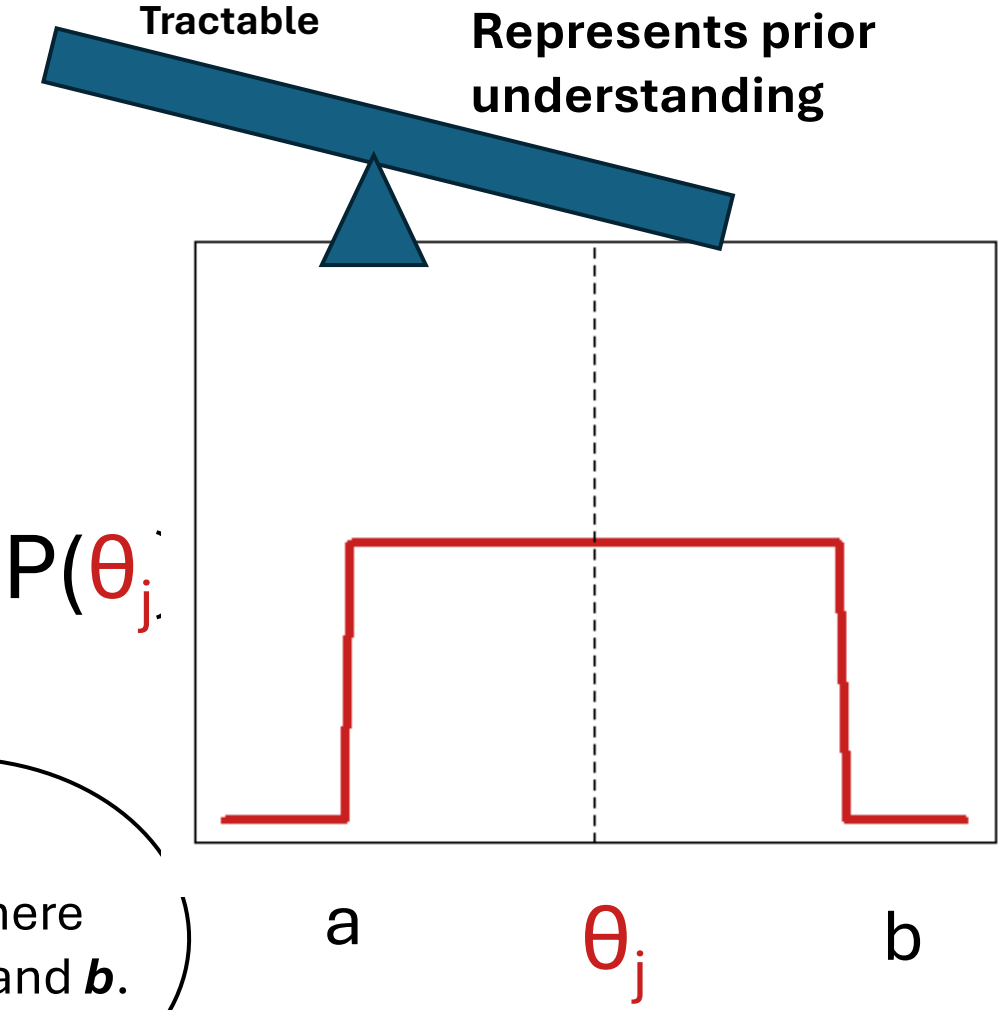
2. Specify priors

3. Sample posterior

$$P(\boldsymbol{\theta}|\boldsymbol{D}) = \frac{P(\boldsymbol{D}|\boldsymbol{\theta})P(\boldsymbol{\theta})}{P(\boldsymbol{D})}$$



It's somewhere between ***a*** and ***b***.



Building our model

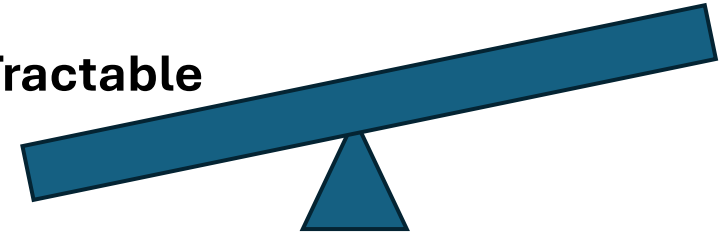


1. Specify the likelihood

Data generating function

2. Specify priors

Tractable



Choose prior functions that will combine with the likelihood function to **create posteriors we can easily sample**.

3. Sample posterior

$$P(\boldsymbol{\theta}|\boldsymbol{D}) = \frac{P(\boldsymbol{D}|\boldsymbol{\theta})P(\boldsymbol{\theta})}{P(\boldsymbol{D})}$$

Parameter generating functions

$P(\boldsymbol{\theta})$

$\theta_j \mid \theta_j^* \sim f(\theta_j^*)$

$P(\boldsymbol{\theta}|\boldsymbol{D})$

$\theta_j \mid \theta_j^*, \boldsymbol{D} \sim f(\theta_j^*, \boldsymbol{D})$

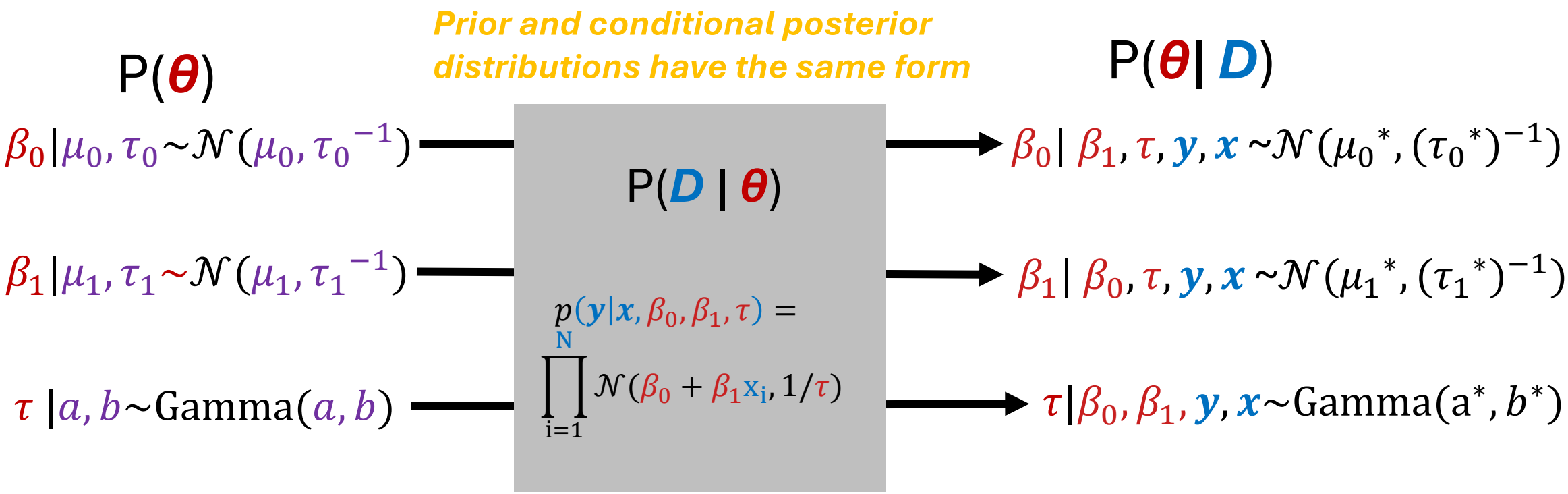
Same type of probability distribution

$\theta_j \sim \mathcal{N}(\mu_j, \tau_j^{-1})$

$\theta_j \mid \boldsymbol{D} \sim \mathcal{N}(\mu *_j, \tau *_j^{-1})$

Conditionally conjugate prior

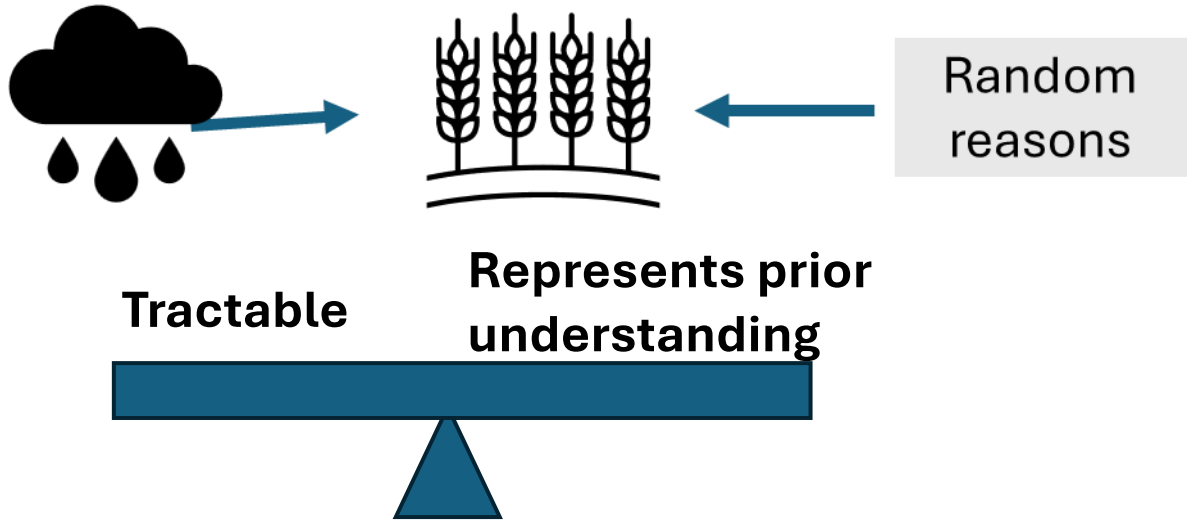
Conditionally conjugate priors for simple linear regression



- Intuitive for Bayesian updating
- Can speed up processing
- Good default if you don't have strong prior understanding

- May not adequately model for prior understanding in all situations
- Not required for Bayesian computation

Building our model



1. Specify the likelihood

2. Specify priors

3. Sample posterior

$$P(\boldsymbol{\theta}|\boldsymbol{D}) = \frac{P(\boldsymbol{D}|\boldsymbol{\theta})P(\boldsymbol{\theta})}{P(\boldsymbol{D})}$$

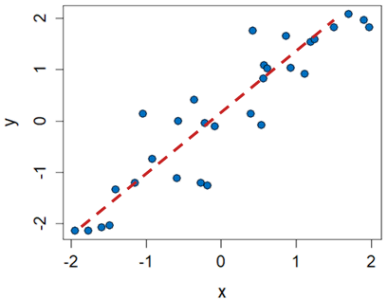
Use **tractable priors** (e.g., conditionally conjugate) and adjust **hyperparameters** to **reflect prior understanding**.

$$P(\theta|D) = \frac{P(D|\theta)P(\theta)}{P(D)}$$

Simple linear regression:

$$y = \beta_0 + \beta_1 x + \varepsilon$$

$$\varepsilon \sim \mathcal{N}(0, 1/\tau)$$



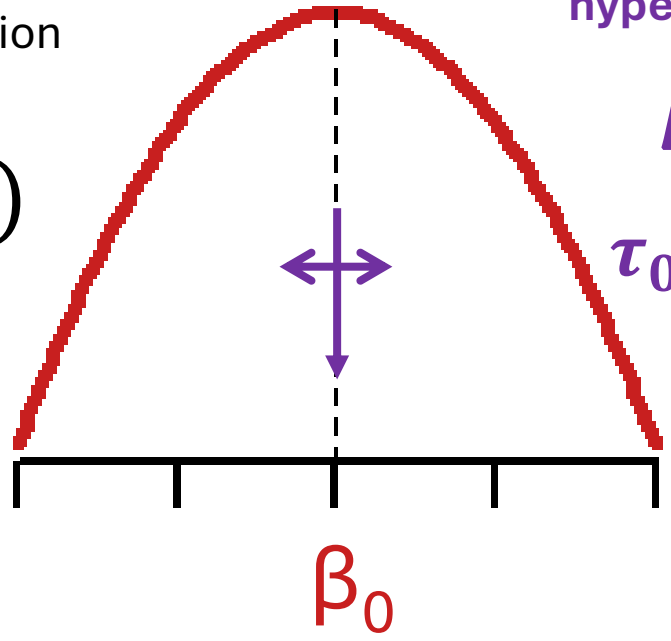
Intercept



Since we standardized **x** and **y**, the intercept is probably close to zero

Conjugate → Normal distribution

$P(\beta_0)$



hyperparameters

$$\mu_0 = 0$$

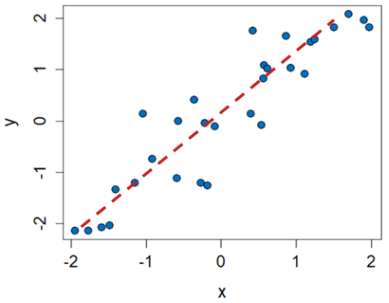
$$\tau_0^{-1} =$$ *Some number expressing our uncertainty*

$$P(\boldsymbol{\theta}|\boldsymbol{D}) = \frac{P(\boldsymbol{D}|\boldsymbol{\theta})P(\boldsymbol{\theta})}{P(\boldsymbol{D})}$$

Simple linear regression:

$$\boldsymbol{y} = \beta_0 + \beta_1 \boldsymbol{x} + \boldsymbol{\varepsilon}$$

$$\boldsymbol{\varepsilon} \sim \mathcal{N}(0, 1/\tau)$$



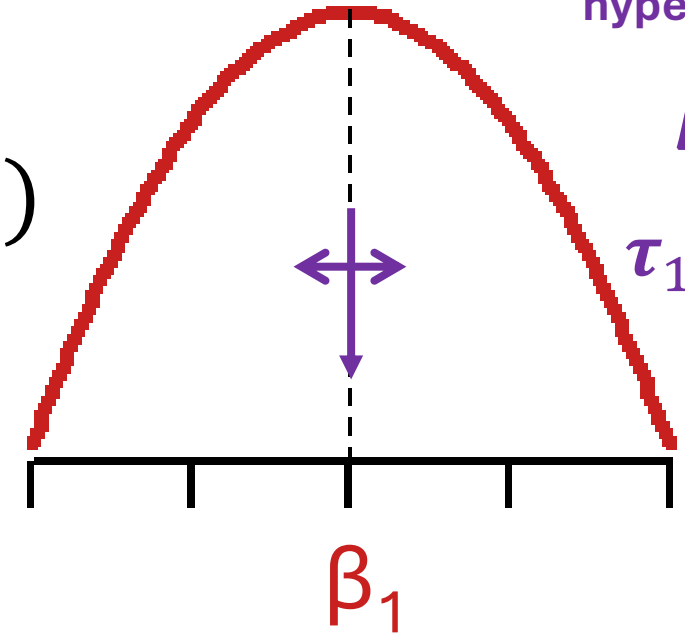
Slope



The slope of **x** vs. **y** could be negative, positive, or zero.

Conjugate → Normal distribution

$P(\beta_1)$



β_1

hyperparameters

$$\mu_1 = 0$$

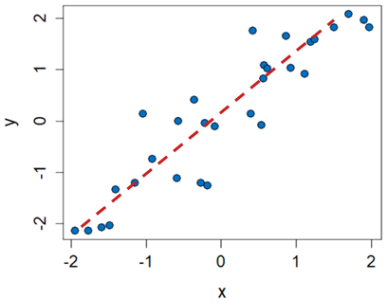
$$\tau_1^{-1} =$$ *Some number expressing our uncertainty*

$$P(\theta|D) = \frac{P(D|\theta)P(\theta)}{P(D)}$$

Simple linear regression:

$$y = \beta_0 + \beta_1 x + \varepsilon$$

$$\varepsilon \sim \mathcal{N}(0, 1/\tau)$$



Precision

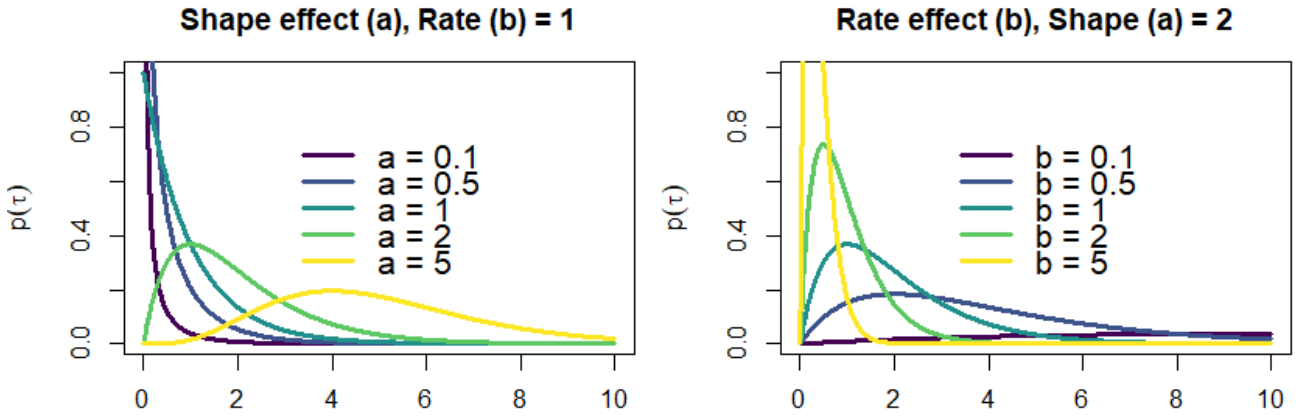


Precision is a number greater than zero. Smaller precision means bigger variance.

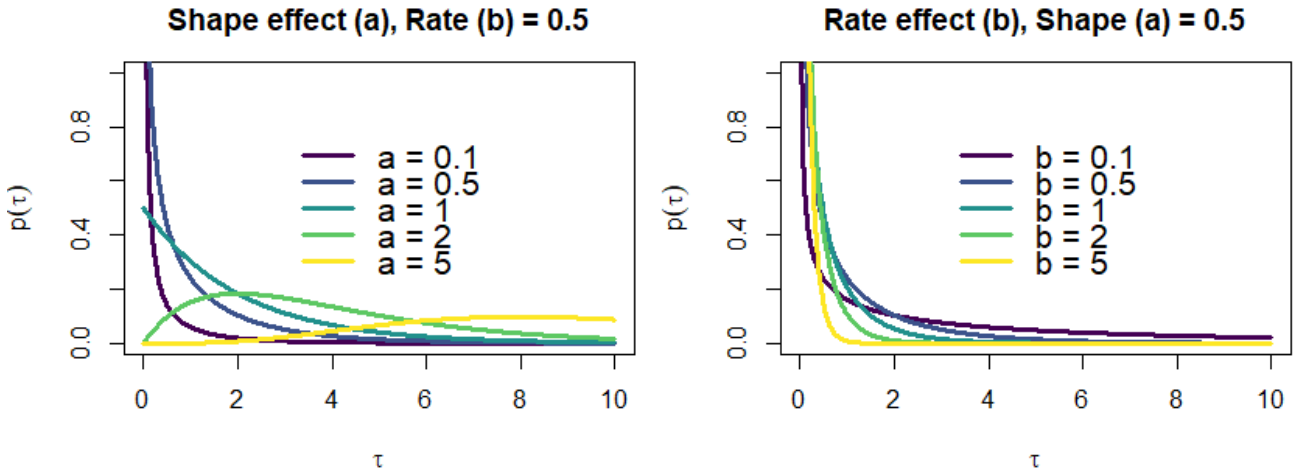
Conjugate → Gamma distribution

a = shape

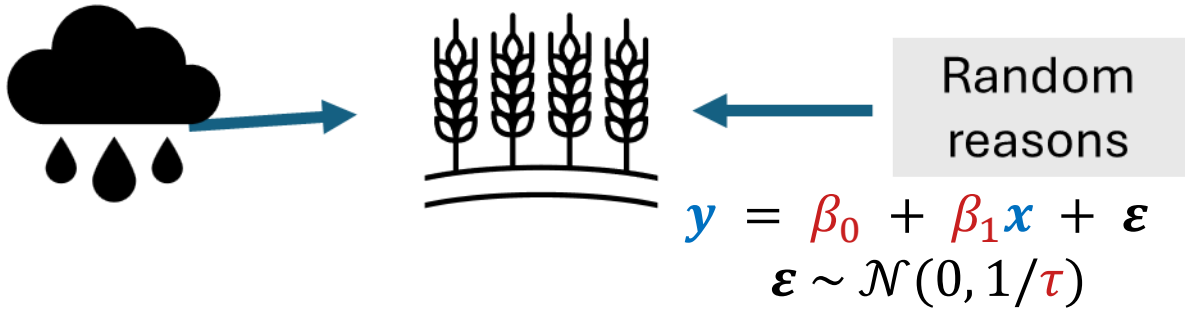
b = rate



Tricky—hyperparameter choices affect posterior sampling



Building our model



1. Specify the likelihood

2. Specify priors

3. Sample posterior

$$P(\theta|D) = \frac{P(D|\theta)P(\theta)}{P(D)}$$

Normalizing constant

Sample **one parameter at a time** conditional on the **data** and **other parameters**

Full conditional posterior

	Likelihood	Prior
$p(\beta_0 \mid \beta_1, \tau, y, x) \propto$	$p(y, x \mid \beta_0, \beta_1, \tau)$	$p(\beta_0)$
$p(\beta_1 \mid \beta_0, \tau, y, x) \propto$	$p(y, x \mid \beta_0, \beta_1, \tau)$	$p(\beta_1)$
$p(\tau \mid \beta_0, \beta_1, y, x) \propto$	$p(y, x \mid \beta_0, \beta_1, \tau)$	$p(\tau)$

Gibbs sampling!

Gibbs sampling

- A **computer algorithm** used to **approximate the full posterior distribution** of one or more parameters

- Works by **iteratively sampling** the **full conditional distributions** of each parameter

- One member of the broader family of **Markov Chain Monte Carlo (MCMC)** methods

To build our Gibbs sampler, we need…

1. Data on rainfall and rice yield.



2. A data-generating model

$$y = \beta_0 + \beta_1 x + \varepsilon$$
$$\varepsilon \sim \mathcal{N}(0, 1/\tau)$$

3. A likelihood function representing our data-generating model

$$p(y, x|\beta_0, \beta_1, \tau) = \prod_{i=1}^{N} \mathcal{N}(\beta_0 + \beta_1 x_i, 1/\tau)$$

4. Prior distributions and hyperparameters for each parameter

intercept $p(\beta_0) = \mathcal{N}(\mu_0, {\tau_0}^{-1})$

slope $p(\beta_1) = \mathcal{N}(\mu_1, {\tau_1}^{-1})$

precision $p(\tau) = \text{Gamma}(a, b)$

Hyperparameters $\mu_0 = 0$ $\tau_0 = 1$
$\mu_1 = 0$ $\tau_1 = 1$
$a = 2$ $b = 1$

5. Full conditional posterior distributions for each parameter → **Calculate using Bayes' Rule!**

Full conditional posteriors via Bayes' Rule

Intercept β_0

Likelihood | Prior

$$p(\beta_0 | \beta_1, \tau, \boldsymbol{y}, \boldsymbol{x}) \propto p(\boldsymbol{y}, \boldsymbol{x} | \beta_0, \beta_1, \tau)\, p(\beta_0)$$

$$= \prod_{i=1}^{N} \mathcal{N}(\beta_0 + \beta_1 x_i, 1/\tau)\, \mathcal{N}(\mu_0, \tau_0^{-1})$$

$$= \prod_{i=1}^{N} \sqrt{\frac{\tau}{2\pi}}\, e^{-\frac{\tau}{2}(y_i - (\beta_0 + \beta_1 x_i))^2} \sqrt{\frac{\tau_0}{2\pi}}\, e^{-\frac{\tau_0}{2}(\beta_0 - \mu_0)^2}$$

Some clever rearranging *

$$\beta_0 | \beta_1, \tau, \boldsymbol{y}, \boldsymbol{x} \sim \mathcal{N}\left(\frac{\tau_0 \mu_0 + \tau \sum_{i=1}^{N} (y_i - \beta_1 x_i)}{\tau_0 + \tau N}, (\tau_0 + \tau)^{-1}\right)$$

<u>Normal probability density function</u>

$$z | \mu, \tau \sim \mathcal{N}(\mu, \tau^{-1})$$

$$p(z | \mu, \tau) = \mathcal{N}(\mu, \tau^{-1})$$

$$p(z | \mu, \tau) = \sqrt{\frac{\tau}{2\pi}}\, e^{-\frac{\tau}{2}(z - \mu)^2}$$

**** Read this for a full breakdown of the steps*** https://stmorse.github.io/journal/gibbs.html

Full conditional posteriors via Bayes' Rule

Slope β_1

$$p(\beta_1 | \beta_0, \tau, \boldsymbol{y}, \boldsymbol{x}) \propto p(\boldsymbol{y}, \boldsymbol{x} | \beta_0, \beta_1, \tau)\, p(\beta_1)$$

$$= \prod_{i=1}^{N} \mathcal{N}(\beta_0 + \beta_1 x_i, 1/\tau)\, \mathcal{N}(\mu_1, {\tau_1}^{-1})$$

$$= \prod_{i=1}^{N} \sqrt{\frac{\tau}{2\pi}}\, e^{-\frac{\tau}{2}(y_i - (\beta_0 + \beta_1 x_i))^2} \sqrt{\frac{\tau_1}{2\pi}}\, e^{-\frac{\tau_1}{2}(\beta_1 - \mu_1)^2}$$

Some clever rearranging *

$$\beta_1 | \beta_0, \tau, \boldsymbol{y}, \boldsymbol{x} \sim \mathcal{N}\left(\frac{\tau_1 \mu_1 + \tau \sum_{i=1}^{N} (y_i - \beta_0) x_i}{\tau_1 + \tau \sum_{i=1}^{N} x_i^2}, (\tau_1 + \tau \sum_{i=1}^{N} x_i^2)^{-1}\right)$$

Normal probability density function

$$z | \mu, \tau \sim \mathcal{N}(\mu, \tau^{-1})$$

$$p(z | \mu, \tau) = \mathcal{N}(\mu, \tau^{-1})$$

$$p(z | \mu, \tau) = \sqrt{\frac{\tau}{2\pi}}\, e^{-\frac{\tau}{2}(z - \mu)^2}$$

**** Read this for a full breakdown of the steps*** https://stmorse.github.io/journal/gibbs.html

Full conditional posteriors via Bayes' Rule

Precision τ

$$p(\tau|\beta_0, \beta_1, \boldsymbol{y}, \boldsymbol{x}) \propto p(\boldsymbol{y}, \boldsymbol{x}|\beta_0, \beta_1, \tau)\, p(\tau)$$

$$= \prod_{i=1}^{N} \mathcal{N}(\beta_0 + \beta_1 x_i, 1/\tau)\, \text{Gamma}(a, b)$$

$$= \prod_{i=1}^{N} \sqrt{\frac{\tau}{2\pi}}\, e^{-\frac{\tau}{2}(y_i - (\beta_0 + \beta_1 x_i))^2}\, \frac{b^a}{\Gamma(a)} \tau^{a-1} e^{-b\tau}$$

Some clever rearranging *

$$\tau|\beta_0, \beta_1, \boldsymbol{y}, \boldsymbol{x} \sim \text{Gamma}\left(a + \frac{N}{2}, b + \frac{1}{2}\sum_{i=1}^{N}(y_i - (\beta_0 + \beta_1 x_i))^2\right)$$

Gamma probability density function

$$x|a, b \sim \text{Gamma}(a, b)$$

$$p(x|a, b) = \text{Gamma}(a, b)$$

$$p(x|a, b) = \frac{b^a}{\Gamma(a)} x^{a-1} e^{-bx}$$

**** Read this for a full breakdown of the steps*** https://stmorse.github.io/journal/gibbs.html

We can code the full conditional posterior for a single parameter as a sampling function in R!

Intercept $$\beta_0 | \beta_1, \tau, \boldsymbol{y}, \boldsymbol{x} \sim \mathcal{N}\left(\frac{\tau_0\mu_0 + \tau\sum_{i=1}^{N}(y_i - \beta_1 x_i)}{\tau_0 + \tau N}, (\tau_0 + \tau)^{-1}\right)$$

```
sample_b0 <- function(x,y, beta_1, tau, mu_0, tau_0){
      N <- length(x)
      new_precision <-  tau_0 + tau * N
      new_mean <- (tau_0 * mu_0 + tau * sum(y - beta_1 * x))/new_precision
      b0samp <- rnorm(1,new_mean, 1/sqrt(new_precision))
      return(b0samp)
}
```

Generates one value of β_0 *by* ***sampling on a normal distribution*** *with mean and sd* ***calculated from input values*** *for the* ***data***, ***parameters***, *and* ***hyperparameters***

We can code the full conditional posterior for a single parameter as a sampling function in R!

Slope $$\beta_1 | \beta_0, \tau, \boldsymbol{y}, \boldsymbol{x} \sim \mathcal{N}\left(\frac{\tau_1\mu_1 + \tau\sum_{i=1}^{N}(y_i - \beta_0)x_i)}{\tau_1 + \tau\sum_{i=1}^{N} x_i^2}, (\tau_1 + \tau\sum_{i=1}^{N} x_i^2)^{-1}\right)$$

```
sample_b1 <- function(x,y, beta_0, tau, mu_1, tau_1){
      N <- length(x)
      new_precision <-  tau_0 + tau * sum(x^2)
      new_mean <- (tau_1 * mu_1 + tau * sum(x*(y - beta_0)))/new_precision
      b1samp <- rnorm(1,new_mean, 1/sqrt(new_precision))
      return(b1samp)
}
```

Generates one value of β_1 *by* ***sampling on a normal distribution*** *with mean and sd* ***calculated from input values*** *for the* ***data***, ***parameters***, *and* ***hyperparameters***

We can code the full conditional posterior for a single parameter as a sampling function in R!

Precision $\tau|\beta_0, \beta_1, \boldsymbol{y}, \boldsymbol{x} \sim \text{Gamma}(a + \frac{N}{2}, b + \frac{1}{2}\sum_{i=1}^{N}(y_i - (\beta_0 + \beta_1 x_i))^2)$

```
sample_tau <- function(x,y, beta_0, beta_1, a, b){
      N <- length(x)
      new_shape <- a + N/2
      residual <- y - beta_0 - beta_1 * x
      new_rate <- b + sum(residual^2)/2
      tausamp <- rgamma(1, shape = new_shape, rate = new_rate)
      return(tausamp)
}
```

Generates one value of τ *by* ***sampling on a normal distribution*** *with shape and rate* ***calculated from input values*** *for the* ***data***, ***parameters***, *and* ***hyperparameters***

Gibbs sampling algorithm

1. Set a starting value for each parameter

Initialization (iteration 0):

$\beta_0^{(0)} = 0\ ,\beta_1^{(0)} = 0\ ,\tau^{(0)} = 1$

2. Sample each parameters using input data and previous values of other parameters

Iteration 1:

Update intercept: $\beta_0^{(1)} = sample_b0\left(\beta_1^{(0)}, \tau^{(0)}, \mathbf{x}, \mathbf{y}\right)$

Update slope: $\beta_1^{(1)} = sample_b1\left(\beta_0^{(1)}, \tau^{(0)}, \mathbf{x}, \mathbf{y}\right)$

Update precision: $\tau^{(1)} = sample_tau\left(\beta_0^{(1)}, \beta_1^{(1)}, \mathbf{x}, \mathbf{y}\right)$

3. Repeat for many iterations

Iteration t:

Update intercept: $\beta_0^{(t)} = sample_b0\left(\beta_1^{(t-1)}, \tau^{(t-1)}, \mathbf{x}, \mathbf{y}\right)$

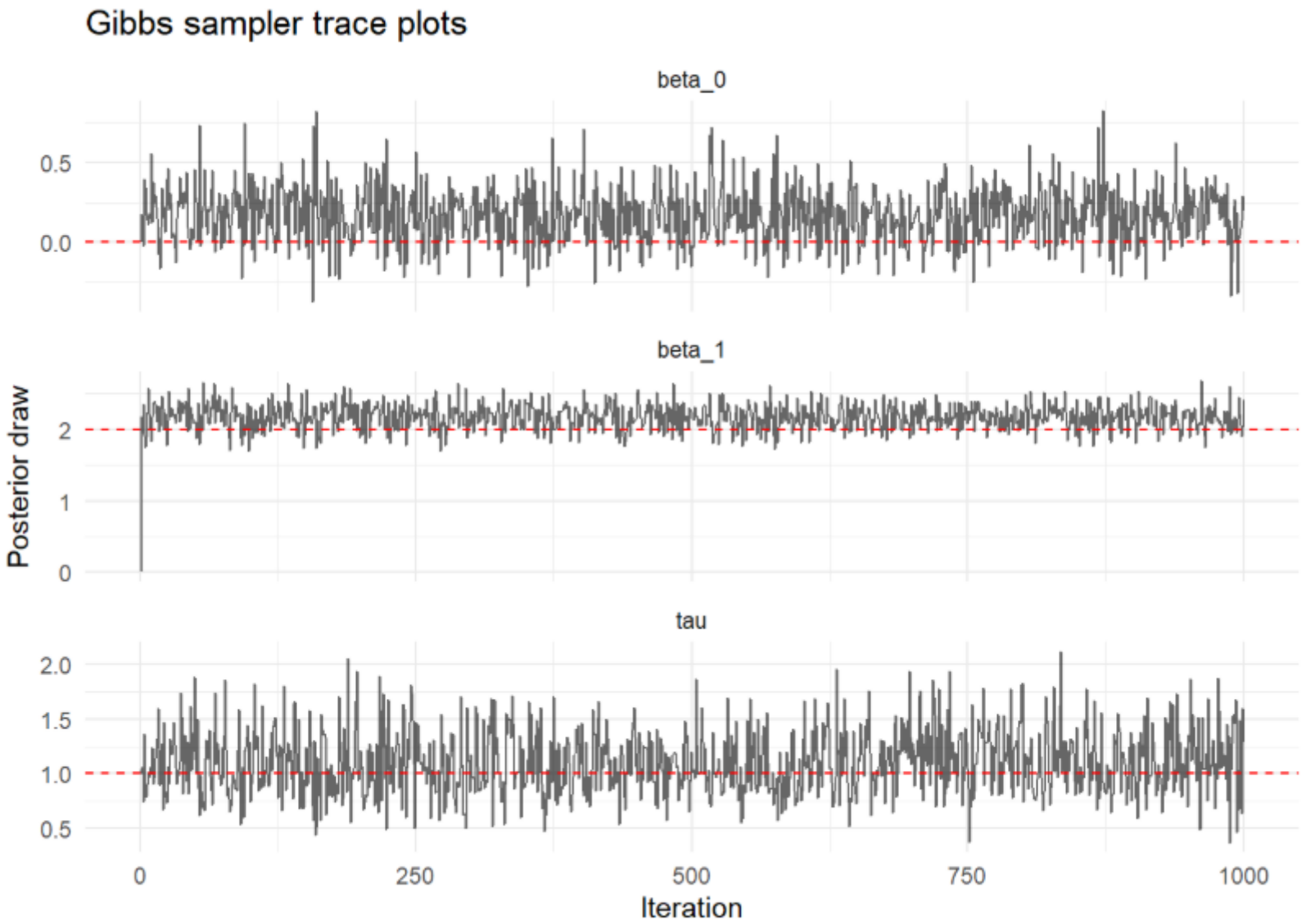
Update slope: $\beta_1^{(t)} = sample_b1\left(\beta_0^{(t)}, \tau^{(t-1)}, \mathbf{x}, \mathbf{y}\right)$

Update precision: $\tau^{(t)} = sample_tau\left(\beta_0^{(t)}, \beta_1^{(t)}, \mathbf{x}, \mathbf{y}\right)$

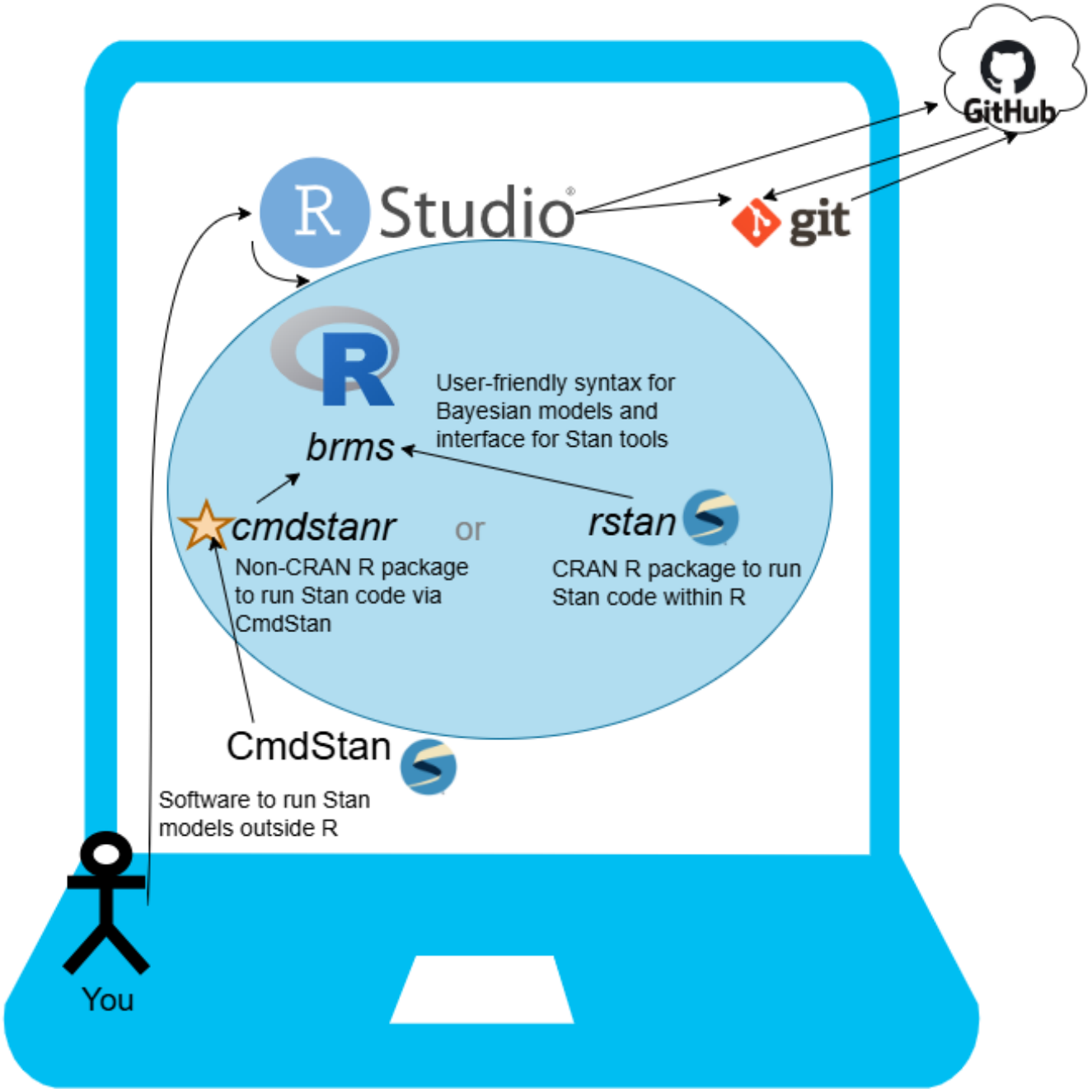
Optional: Look through the Gibbs sampler code and exercises in R.

Runnable code:
GibbsSampler.Rmd

Formatted output:
GibbsSampler.html



Bayesian software tools



Bayesian tools

Stan  https://mc-stan.org/

- Software and programming language for Bayesian data analysis
- Free and open-source
- Can be used in multiple programming environments (**R**, Python, Julia, Unix, etc.)

$$y_n \sim \text{normal}(\alpha + \beta X_n, \sigma).$$

```
data {
  int<lower=0> N;
  vector[N] x;
  vector[N] y;
}
parameters {
  real alpha;
  real beta;
  real<lower=0> sigma;
}
model {
  y ~ normal(alpha + beta * x, sigma);
}
```

Stan syntax: C++

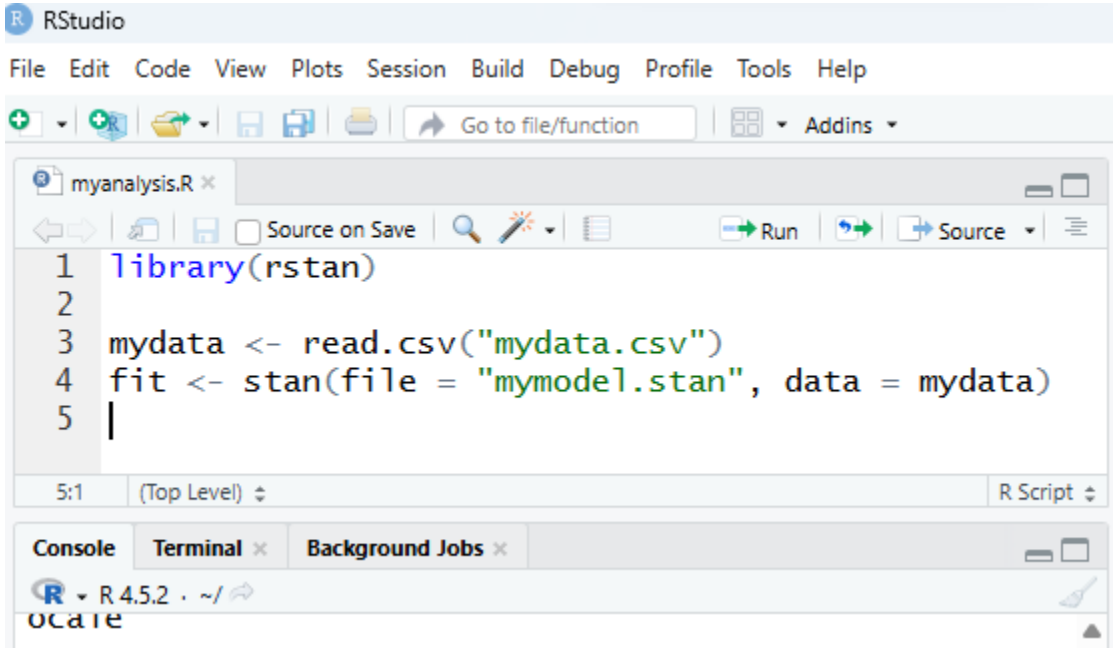
Bayesian tools

- R package
- Runs stan code inside R
- Uses a C++ interpreter (RTools)

mymodel.stan

```
data {
  int<lower=0> N;
  vector[N] x;
  vector[N] y;
}
parameters {
  real alpha;
  real beta;
  real<lower=0> sigma;
}
model {
  y ~ normal(alpha + beta * x, sigma);
}
```

RStan



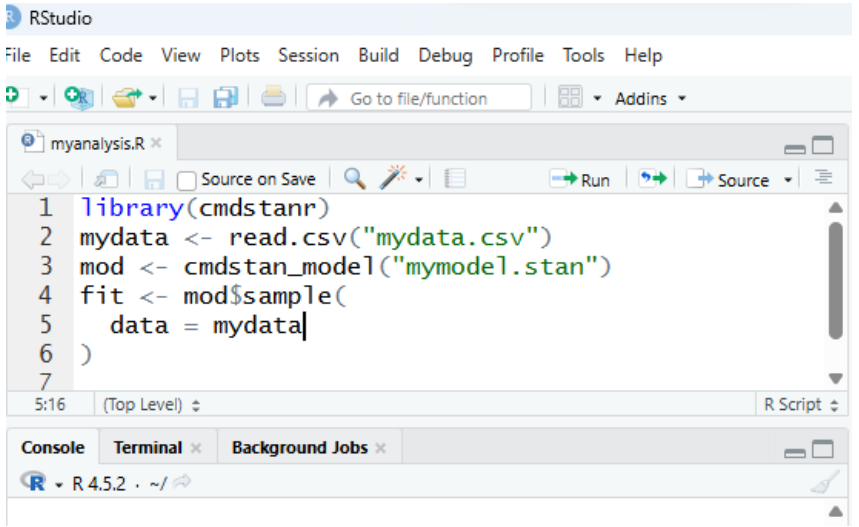
Bayesian tools

cmdstanr

- R package
- Alternative to Rstan (also requires RTools)
- Bayesian computation done in an external software, CmdStan

mymodel.stan

```
data {
  int<lower=0> N;
  vector[N] x;
  vector[N] y;
}
parameters {
  real alpha;
  real beta;
  real<lower=0> sigma;
}
model {
  y ~ normal(alpha + beta * x, sigma);
}
```



rstan vs cmdstanr

Older, used in lots of help docs and examples

Newer, becoming more widely used

Managed by CRAN

Outside of CRAN, so better able to keep up with new Stan releases

Less memory overhead and less prone to crashing

Bayesian tools brms

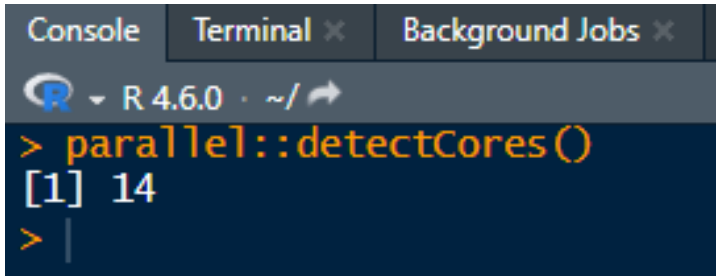
- R package for Bayesian data workflow
- Designed for complex multilevel models
- Translates familiar R syntax into Stan code
- Works with rstan or cmdstanr to run models
- Integrates with other Bayesian packages

Hardware requirements

MCMC sampling can take minutes to hours depending on the dataset size and model complexity

Personal computer

- Usually sufficient for small-medium datasets (e.g., a field experiment) and/or simpler models.
- Computer should have at least 4 spare processor cores for parallel MCMC chains



High Performance Computing Cluster (HPCC)

- Helpful for larger datasets/more complex models
- Available at most major universities and research institutions.
- You connect to a remote server from your personal computer.
- You run models with shell scripting (SLURM) or an interactive R Studio environment.
- Requires copying data files and scripts to the remote server (good file organization is essential!)

OPEN OnDemand

OnDemand provides an integrated, single access point for all of your HPC resources.

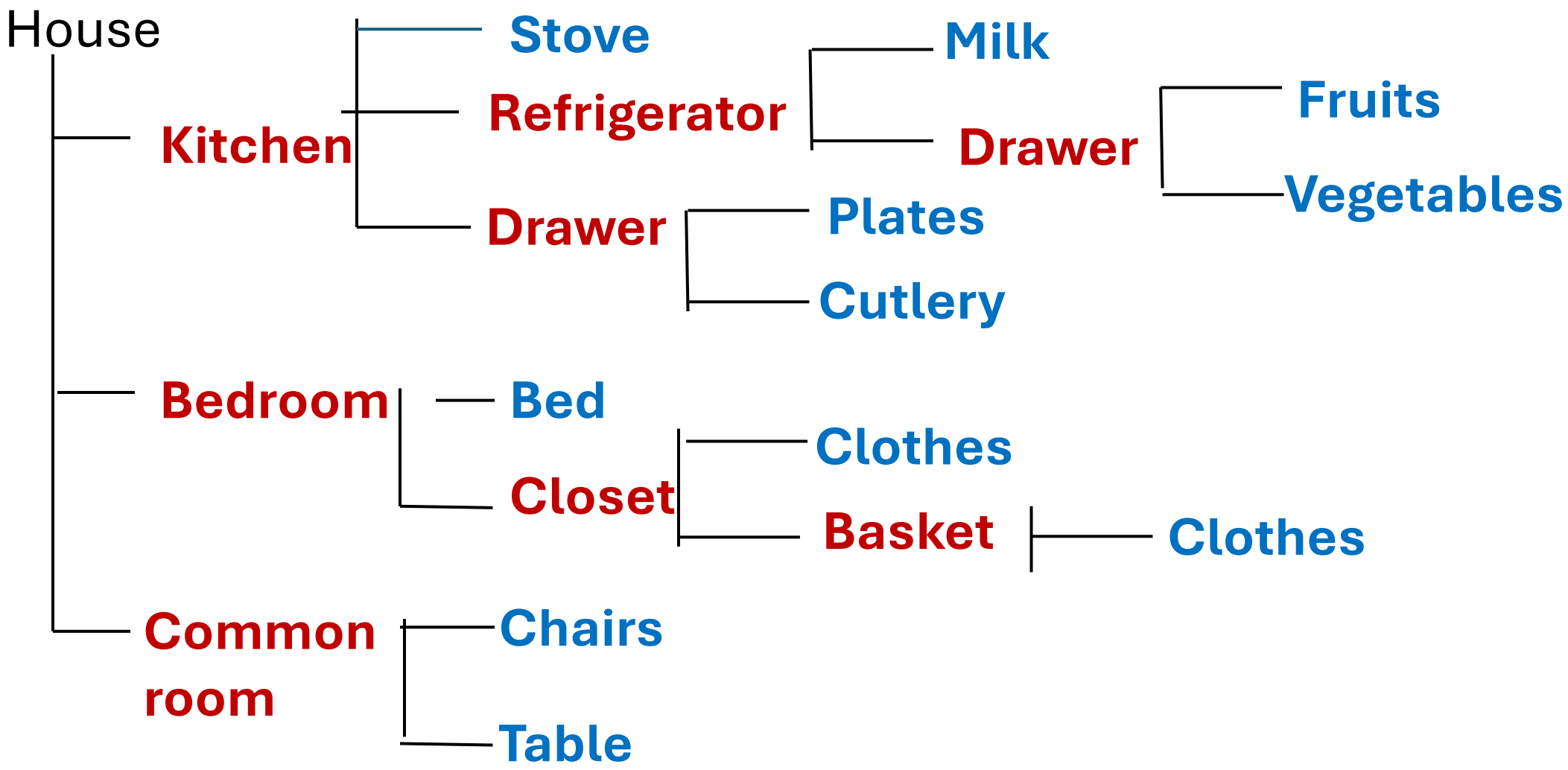
USDA SCINet Scientific Computing
U.S. DEPARTMENT OF AGRICULTURE

An aside about file organization...

File organization is a barrier to computational data analysis

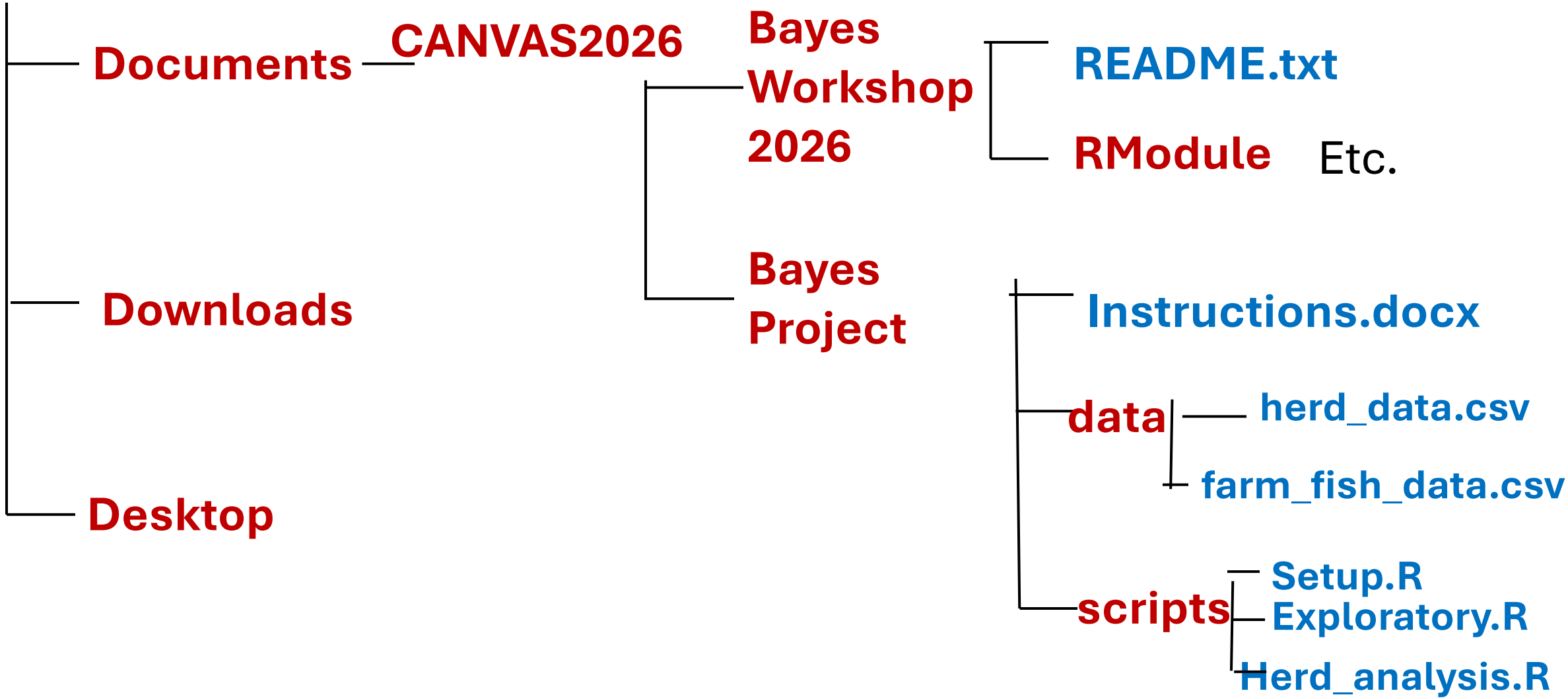
- To analyze data in R, you need to tell R where the data “live”
- In a data analysis project, you will use and create many files (including many generated in R).
- Having a system for organizing your files AT THE BEGINNING will make your life easier.

Organize the files in your computer like the objects in your house



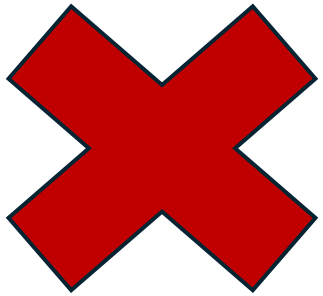
Organize the files in your computer like the objects in your house

Your computer hard drive: **Directories = Compartments**, **Files = Objects**



Use informative file and folder names

Analysis.R



CANVAS2026herdAnalysis v01.R



Add documentation as you go

